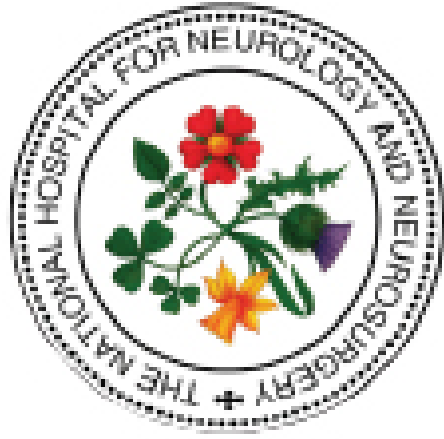




The National Hospital for Neurology and Neurosurgery Neuroscience Critical Care Unit (NCCU)

3. DIAGNOSING DEATH by NEUROLOGICAL CRITERIA (DNC)

Academy of Medical Royal Colleges (2025) Code of Practice in UK

MAINTAINING THE PATIENT FOR TESTING / POST-COMPLETION OF TESTS

1. NEUROLOGY

- **GCS 3:** No eye opening No verbal response No motor response
- **Pupils fixed:** do not respond to sharp changes in intensity of incident light either directly or consensually
- No other cranial nerve function (*see Tests for Absence of Brainstem Function*)

2. RESPIRATORY

- No spontaneous respiration
- Absence of gag reflex on oral suction
- Absence of cough on tracheal suction
- Normal pH (7.35-7.45) and acid base balance
- Avoid hypoxia ($\text{PaO}_2 < 10\text{kPa}$) if possible
- Normalise CO_2 – *once no longer a need to protect the brain, ventilation parameters may be changed for lung protection in the event that the patient becomes an organ donor (see Organ Donor Protocol)*
- Allow CO_2 to rise to $\geq 5.3\text{kPa}$ in preparation for testing

3. CARDIOVASCULAR and METABOLIC* AT TIME OF TESTING

- Normal (or patient baseline) intensive care physiological parameters targeted prior to / during testing
- MABP consistently $\geq 60\text{mmHg}$ (*vasopressor - noradrenaline / vasopressin*)
If noradrenaline requirements high start vasopressin IV (bolus 0.5 units and then infusion 0.5-2units/h – see protocol)
- Minimum core temperature 36°C
- Serum sodium (Na^+) $125-160\text{mmol/L}$ (inclusive) **
- Serum potassium (K^+) $> 2\text{mmol/L}$
- Serum magnesium (Mg^{2+}) $0.5-3.0\text{mmol/L}$ (inclusive)
- Serum Phosphate (PO_4^{3-}) $0.5-3.0\text{mmol/L}$ (inclusive)
- Blood glucose $3-20\text{mmol/L}$ (inclusive) - *checked prior to each test - no longer a need to protect brain therefore tighter range not required*
- Normal pH 7.35-7.45

* It is recognised that circulatory, electrolyte, metabolic, endocrine disturbances (hypotension, hypernatraemia, diabetes insipidus) are likely accompaniments to both permanent cessation of brainstem function and devastating brain injury.

Such disturbances do not preclude application of neurological criteria. However, if these cannot be corrected and judged to be potentially contributing to cessation of brainstem function, ancillary investigation must be considered.

** Na^+ levels $> 160\text{mmol/L}$ can be associated with unresponsiveness and should be considered if primary cause of coma prior to testing is uncertain – if clear that Na^+ levels $> 160\text{mmol/L}$ are **effect** rather than **cause** of cessation of brainstem function then BSDT can be completed

4. SEDATION

- Sedatives, analgesics, muscle relaxants discontinued for $> 6\text{h}$ prior to testing (*longer if midazolam and thiopentone used*)
- Reversing drugs (e.g. naloxone, flumazenil) / peripheral nerve stimulator recommended if NMBAs have been used

5. HYDRATION

- IV fluids to maintain MABP $\geq 60\text{mmHg}$
- Continue NG feeding even if likely to be only for short period (*important if patient becomes organ donor – rate may need to be reduced if absorption impaired*)
- NG water / 5% dextrose if needed to maintain $\text{Na}^+ \leq 160\text{mmol/L}$

6. ELIMINATION

- DI should be suspected if urine output $> 100\text{mL/h}$ & Na^+ rising
- DDAVP 0.4mcg if evidence of diabetes insipidus to control electrolyte balance and volume status - *should be given if urine output high (even if not reaching 2L/4h) and Na^+ climbing ($> 160\text{mmol/L}$)*
 - Consider IV vasopressin infusion if requiring frequent boluses

7. PREPARATION for TESTING

1. Evidence of **PERMANENT BRAIN DAMAGE of KNOWN AETIOLOGY**
2. Case records, past medical history (GP contacted if required), relevant imaging should all be available
3. **EXCLUSION of REVERSIBLE CAUSES of COMA and APNOEA**
4. Standard ICU cardio-respiratory monitoring (to ensure haemodynamic stability), medication record and history, blood and urine drug assay results (where relevant), drug antagonists (e.g. flumazenil, naloxone), peripheral nerve stimulator, recent serum glucose and biochemistry, thermometer, patient warming device if required
5. Tests for Absence of Brainstem Function: *Brainstem reflexes*
 - Bright light source
 - Small gauze sterile swabs
 - Otoscope with disposable earpieces
 - 50mL luer lock syringe and disposable quill
 - Ice-cold water
 - Spatula
 - Yankauer sucker or laryngoscope
 - Endotracheal suction catheters

Apnoea test

 - Haemodynamic monitoring (continuous ECG, arterial line)
 - ABG analysis including blood gas syringes x 4
 - Pulse oximetry and end-tidal CO_2 monitoring
 - Means of delivering oxygen to trachea by bulk flow (*Mapleson C Waters Circuit which allows CPAP or endotracheal suction catheter and oxygen tubing*)

POST-COMPLETION OF TESTS

If brainstem death tests (BSDT) show that patient is brainstem dead:

1. All neuromonitoring (invasive and non-invasive) should be discontinued – this includes GCS, pupil / limb assessments, RASS, removal of intracranial devices

This applies to all patients who are Diagnosed Dead by Neurological Criteria

2. If the patient is **not** to become an organ donor:
 - All cardiovascular and respiratory observations should cease
 - Monitoring of fluid intake / output should cease
 - EPIC entries immediately transfer to 'Care following death' documentation

Only exception is if family do not accept validity of BSDT – in this case only neuromonitoring should be discontinued as in 1. above while awaiting court decision. Legal advice should be sought urgently if agreement cannot be reached in a reasonable amount of time and at most within a few days

3. If the patient is **to proceed** to organ donation:
 - Continue all cardiovascular and respiratory observations
 - Continue monitoring of fluid intake / output
 - Follow 'Management of patient for organ donation' with input from SNOD

LIMB and TRUNK MOVEMENTS

The significance of any reflex movements that may occur should be explained to relatives and staff to enable them to understand that they do not involve conscious voluntary movements

- After confirming death using neurological criteria, non-brain mediated movements are well recognised and can include spontaneous or stimulated spinal reflexes, automatisms (occasionally complex) and muscle fasciculations
- These movements are independent of the brain and are initiated at the spinal cord or at a neuromuscular level and may appear or persist while mechanical ventilation and other intensive care interventions maintain oxygenated blood to the spinal cord and other parts of the body
- The longer somatic support is continued after diagnosis and confirmation of death using neurological criteria the more likely non-brain mediated movements will appear and potentially the more exaggerated, they will be
- Such movements do not indicate any form of consciousness associated with human life nor any brainstem function

DOCUMENTATION

Documented **time of death** is the time of **2nd set of brainstem tests** (or time of ancillary testing if this occurs after BSDT) when diagnosis of brain death is confirmed, and not when mechanical ventilation is discontinued

Brainstem Death Checklist completed by team carrying out BSDT

SWITCHING off the VENTILATOR

If patient **not** to be an organ donor, when appropriate time given to relatives, the ventilator should be switched off and all infusions discontinued

There should be no 'weaning' process since patient has been certified dead, as this can be confusing for both family and staff members

There should be no further 'confirmation of death' after the ventilator has been switched off

Continued mechanical ventilation and other intensive care interventions may maintain (some) organ function in the body after death, but they are inappropriate beyond a short interval that may be justified as part of end of life and bereavement care.